

1. Multiple Regression Equation

a. Find the least-squares regression equation of X_3 on X_1 and X_2

The general form is $X_3 = a + b_1X_1 + b_2X_2$. Using the method of least squares to solve for the coefficients:

1. Calculations:

- $\sum X_1 = 48, \sum X_2 = 42, \sum X_3 = 300$
- $\sum X_1^2 = 474, \sum X_2^2 = 454, \sum X_1X_2 = 237$
- $\sum X_1X_3 = 1860, \sum X_2X_3 = 2694$

2. Solving the Normal Equations:

- $300 = 6a + 48b_1 + 42b_2$
- $1860 = 48a + 474b_1 + 237b_2$
- $2694 = 42a + 237b_1 + 454b_2$

3. Resulting Coefficients: $a \approx 97.58, b_1 \approx -3.37, b_2 \approx -5.34$

- **Regression Equation:**
 $X_3 = 97.58 - 3.37X_1 - 5.34X_2$

b. Estimate x_3 when $x_1 = 10$ and $x_2 = 6$

- $X_3 = 97.58 - 3.37(10) - 5.34(6)$
- $X_3 = 97.58 - 33.7 - 32.04 = 31.84$

2. Standard Error of Estimate

- **Formula:** $S_{3.12} = \sqrt{\frac{\sum(X_3 - \hat{X}_3)^2}{n-k-1}}$
- **Result:** $S_{3.12} \approx 4.65$

3. Coefficient of Multiple Correlation (R)

- $R = \sqrt{1 - \frac{SSE}{SST}}$
 - $SST = \sum(X_3 - \bar{X}_3)^2 = 3888$
 - $SSE = 64.86$
 - $R = \sqrt{1 - \frac{64.86}{3888}} = \sqrt{0.9833}$
 - $R = 0.991$
- **Interpretation:** A very strong positive relationship exists between the predictors and X_3 .

4. Partial Correlation ($r_{12.3}$)

1. Zero-order correlations:

- $r_{12} = -0.835$
- $r_{13} = -0.963$
- $r_{23} = 0.941$

2. **Formula:** $r_{12.3} = \frac{r_{12} - (r_{13} \cdot r_{23})}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$

3. Calculation:

- **Numerator:** $-0.835 - (-0.963 \cdot 0.941) = -0.835 + 0.906$
- **Denominator:** $\sqrt{(1 - 0.927)(1 - 0.885)} = \sqrt{0.073 \cdot 0.115}$

4. **Result:** $r_{12.3} \approx 0.775$

5. Real-World Multiple Regression (Housing)